

optimization problem of the type:

Lagrangian

constraint :

Exercise 1

 $y_1, \dots, y_N \in \mathbb{R}$: label
$$\phi: \mathbb{R}^d \rightarrow \mathbb{R}^h \quad : \text{feature map}$$

(a) because $\xi_i = W^T \phi(X_i) - y_i$ and

$$\min_{\mathcal{E}, W} \sum_{i=1}^N \frac{1}{2} \mathcal{E}_i^2$$

we get $g_i(z, w) = z_i - w^T \phi(x_i) + y_i = 0$

therefore rewrite the regularization in form

$$h(w) = \frac{1}{2} \|w\|^2 - C \leq 0$$
$$L(\varepsilon, \omega, \beta, \lambda) = J(\varepsilon, \omega) + \sum_{i=1}^m \beta_i g_i(\varepsilon, \omega) + \sum_{i=1}^2 \lambda_i h_i(\theta)$$

therefore

KKT conditions :

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12/16/1914

Stationarity (1/1/1980)

$$\forall_{i=1}^N, \lambda_i \geq 0$$

dual feasibility (2) (1)

$$\forall_{i=1}^N, \lambda: h_i(\theta) = 0$$

complementary slackness ^(2/)
~~(3/)~~

$$\lambda \cdot \left(\frac{1}{2} \|w\|^2 - C \right) = 0$$

(No)

6/21/1870 171/1870-1870

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13) ~~Static~~ Stationarity

$$\frac{\partial L}{\partial \theta} = 0 \Rightarrow \frac{\partial L}{\partial q_i} = \xi_i + \lambda_i = 0$$

$$\frac{\partial L}{\partial w} = 0 \Rightarrow$$

$$\sum_{i=1}^N \{ \psi(-w^T \phi(x_i)) \} + \lambda w = 0$$

$$\Rightarrow w = \frac{1}{\lambda} \cdot \sum_{i=1}^N f_i \phi(x_i)$$

$$z_i - w^T \phi(x_i) + y_i = 0$$

Task a(b)

$$\frac{\partial L}{\partial \xi_i} = \xi_i + \beta_i = 0 \Rightarrow \xi_i = -\beta_i$$

$$w = \frac{1}{\lambda} \sum_{i=1}^N \beta_i \phi(x_i)$$

plug in $\xi_i = -\beta_i$

$$\xi_i - w^T \phi(x_i) + y_i = 0$$

$$-\beta_i - w^T \phi(x_i) + y_i = 0$$

$$y_i = \beta_i + w^T \phi(x_i)$$

$$= \beta_i + \left(\frac{1}{\lambda} \cdot \sum_{j=1}^N \beta_j \phi(x_j) \right)^T \cdot \phi(x_i)$$

$$= \beta_i + \frac{1}{\lambda} \sum_{j=1}^N \beta_j \phi(x_j)^T \phi(x_i)$$

$$= \beta_i + \frac{1}{\lambda} \sum_{j=1}^N \beta_j \cdot K(x_j, x_i)$$

$$K \in \mathbb{R}^{N \times N}, K_{ji} = K(x_j, x_i)$$

~~write as~~

$$\beta = (\beta_1, \dots, \beta_N)^T, y = (y_1, \dots, y_N)^T$$

$$(w^T \phi(x_1), \dots, w^T \phi(x_N))^T = \frac{1}{\lambda} \cdot K^T \beta$$

~~$y_i = \beta_i + \frac{1}{\lambda} \sum_{j=1}^N \beta_j K_{ji}$~~

$$y = \beta + \frac{1}{\lambda} K \beta$$

$$\lambda y = \lambda \beta + K \beta = (\lambda I + K) \beta$$

$$\Rightarrow \beta = (K + \lambda I)^{-1} \lambda y$$

(c)

$$f(x) = w^T \phi(x)$$

$$w = \frac{1}{\lambda} \sum_{i=1}^N \beta_i \phi(x_i)$$

in terms of the dual parameter β_i :

$$f(x) = \left(\frac{1}{\lambda} \sum_{i=1}^N \beta_i \phi(x_i) \right)^T \phi(x)$$

$$= \frac{1}{\lambda} \sum_{i=1}^N \beta_i K(x_i, x)$$

~~write as~~